HEALTHCARE DIGITAL TWIN • SUMMIT 2026

PULSE_TWIN AI

A Personalized Digital Twin for Proactive Diabetes Management

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Proof of Concept • [Submission Date]

The problem: diabetes management is continuous

Clinical challenge

- **Changing glucose:** levels vary with meals, activity, medication, stress and sleep.
- **Limited visibility:** periodic measurements may miss short-term changes between checks.
- **Reactive decisions:** people and care teams need understandable signals before glucose moves out of range.
- **Individual differences:** the same meal or activity can affect different people differently.

The opportunity

Turn available patient and wearable data into a patient-specific, time-aware view.

The design goal

Explore likely near-term glucose trends and scenarios—not replace clinical judgment.

Target use case

A research/clinical-support prototype for Type 2 diabetes glycemic management.

Our proposed solution

A data-informed patient twin connects context, prediction and scenario exploration.

01

Build the profile

Patient parameters + longitudinal records

02

Observe current state

CGM / wearable signals + recent context

03

Forecast & flag

30-minute glucose prediction + threshold alerts

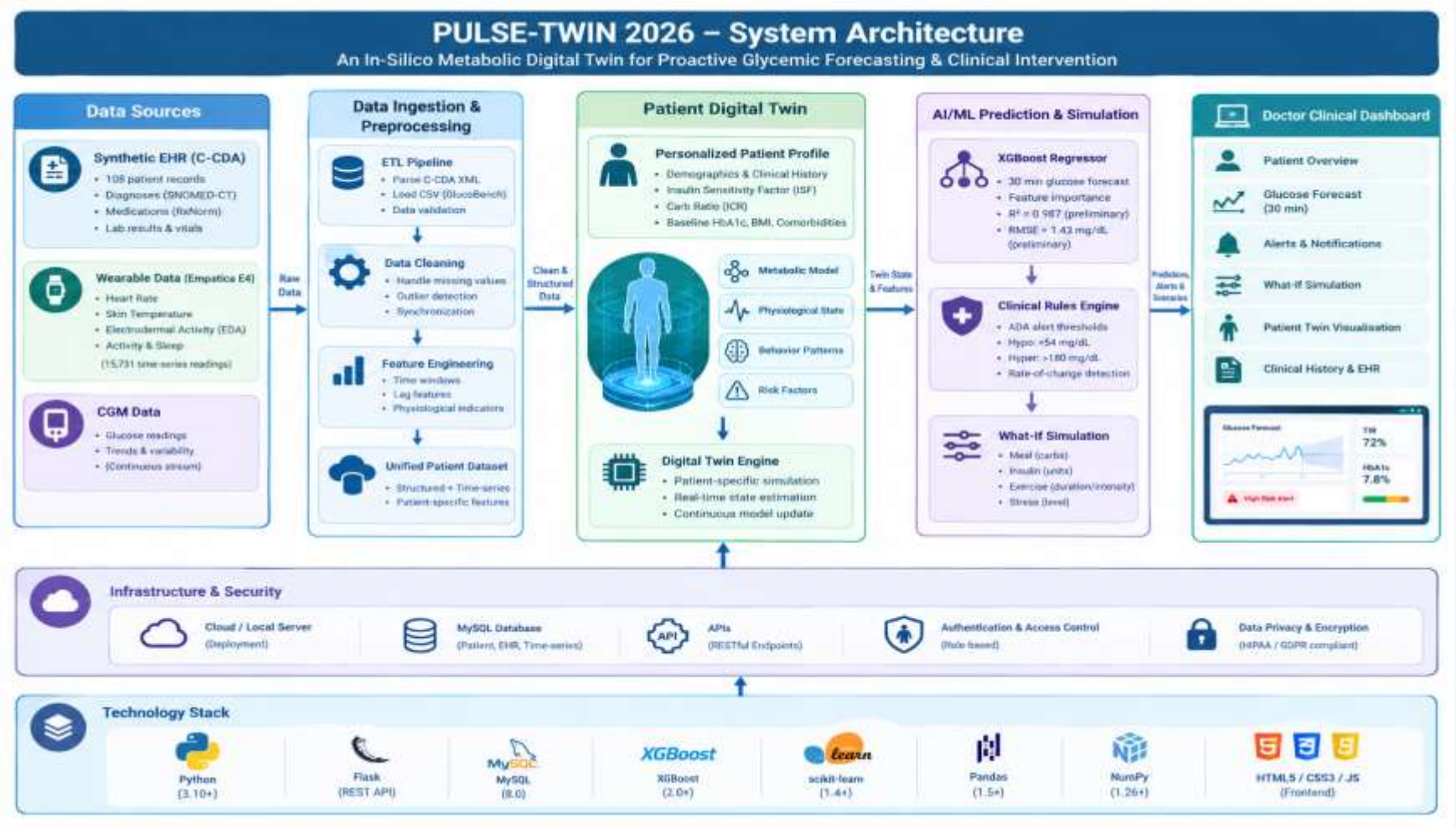
04

Explore scenarios

Meal, insulin, exercise and stress what-ifs

Expected value: earlier awareness, clearer context and safer evaluation of possible responses.

System Architecture



Data sources and ingestion

Synthetic EHR / C-CDA

108 synthetic patient records

3,517 diagnosis entries (SNOMED CT)

3,841 prescription entries (RxNorm)

Wearable / glucose data

15,731 CGM + Empatica E4 readings

Across 10 subjects (as currently reported)

Time-series context for the prototype

Import pipeline

synthea_importer.py

Normalize codes and timestamps

Link records to synthetic subjects

Data quality checks to document

- **Provenance:** identify source, synthetic status, collection method and permitted use.
- **Time alignment:** check timezone, duplicate timestamps, sampling gaps and sensor delays.
- **Missingness:** report absent readings and how missing values are handled.

The patient-specific digital twin

Profile inputs

- **ISF:** insulin sensitivity factor, if defined and available.
- **ICR:** insulin-to-carbohydrate ratio, if defined and available.
- **Baseline HbA1c:** longer-term context, not a short-term glucose forecast.
- **Clinical context:** diagnoses, medication records and available time-series history.
- **Current state:** recent CGM readings and relevant recent events.

Twin builder

twin_builder.py creates or updates a profile from available records.

Schema

schema.sql defines how profile and associated data are stored.

Key limitation

A profile is not automatically a validated physiological simulator; parameter provenance and calibration matter.

AI/ML: short-horizon glucose prediction

Model

XGBoost regression

Implemented in predictor.py

Forecast target

Glucose value 30 minutes ahead

Confirm exact label and units

Outputs

Forecast curve/value + context

Uncertainty should be added

Validation plan

- Split by subject and/or chronological period.
- Fit preprocessing on training data only.
- Compare with simple baselines (last value, persistence).
- Report MAE/RMSE and calibration across relevant ranges.

Important result caveat

Preliminary reported metrics: $R^2 = 0.987$ and RMSE = 1.43 mg/dL. Include these only if reproduced from a leakage-checked held-out test set; otherwise label them preliminary and explain the validation still needed.

What-if scenario simulation

Explore how different assumptions could change the predicted glucose trajectory.

Meal

Carbohydrate amount / timing

Explore a possible post-meal response

Insulin

Scenario input, only if supported

Never present output as dosing advice

Exercise

Activity timing / intensity

Explore a hypothetical activity effect

Stress

Stress scenario parameter

Show assumptions and uncertainty

Interactive doctor dashboard

PROTOTYPE DEMO

Live walkthrough: demonstrate what is implemented today.

01	**Select synthetic patient**	Confirm identity is synthetic and data is loaded.
02	**Review the current state**	Show recent glucose readings and timestamp.
03	**Open the prediction**	Explain the 30-minute forecast and its limitations.
04	**Inspect an alert**	Show threshold, data freshness and relevant context.
05	**Run one scenario**	Change one variable and compare baseline vs simulated output.

Technical stack and implementation

Data ingestion

synthea_importer.py

Imports synthetic C-CDA records and maps diagnosis/prescription codes.

Data model

schema.sql

Defines the storage structure for the twin and related data.

Twin construction

twin_builder.py

Creates the patient-specific profile from available data.

Prediction

predictor.py + XGBoost

Generates the short-horizon glucose forecast.

Simulation

simulation_service.py

Runs what-if scenario logic.

API / frontend

/api/twins/<id>/tick • index.html

Serves live simulation updates and the dashboard.

Impact and future roadmap

Now • PoC Demonstrate ingestion, patient profile, forecast, alerts and what-if workflow.

Next • Technical validation Leakage checks, subject-wise splits, baseline models, error analysis and uncertainty.

Next • Clinical feedback Review use cases, alert thresholds, usability and safe wording with qualified clinicians.

Later • Interoperability Explore FHIR integration, access controls, auditability and deployment governance.

Evaluation and outcomes

Prototype coverage

Data import • patient twin • forecast • alert logic • what-if scenarios

Reported model result

R^2 0.987 / RMSE 1.43 mg/dL — preliminary until independently verified

Evidence to show

Held-out test design, sample size, baseline comparison and error distribution

Recommended evaluation table

Measure	What it tells us	Result	Evidence / notes
MAE / RMSE	Typical prediction error	[Insert verified]	Held-out data split
Baseline comparison	Value beyond persistence	[Insert verified]	Same test set
Coverage / missingness	Data reliability	[Insert verified]	Per subject / sensor

Conclusion

GlucoTwin AI connects patient context, short-horizon prediction and scenario exploration in one prototype.

What we built

A working proof of concept for a personalized glucose-management twin.

What we learned

Data quality, honest evaluation and safe communication are as important as model accuracy.

What comes next

Stronger validation, clinical feedback and responsible integration.

Thank you

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Demo URL: [Insert link] • Open-source license: [Insert exact license]